

GOSHEN, IN, USA

WMO: 724388

Lat: 41.525N

Lon: 85.791W

Elev: 830

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.4	(c) 4.3	(d) -7.6	(e) 3.8	(f) -0.2	(g) -2.4	(h) 5.0	(i) 5.5	(j) 26.6	(k) 25.9	(l) 24.7	(m) 25.3	(n) 7.2	(o) 250	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	90.2	73.6	87.6	72.6	84.6	70.9	77.0	85.7	75.3	83.5	73.7	81.4	11.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	132.1	82.2	72.6	124.4	80.0	71.4	119.4	78.7	40.9	85.3	39.2	83.8	37.7	81.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.9	19.0	-8.4	93.6	6.3	3.4	-12.9	96.1	-16.6	98.0	-20.1	99.9	-24.7	102.4
			-8.9	79.3	6.2	2.0	-13.3	80.7	-16.9	81.9	-20.4	83.0	-24.9	84.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	24.8	27.0	37.8	49.8	60.6	69.7	72.5	70.7	64.6	52.5	41.0	29.9
	DBStd	19.24	11.66	10.95	11.08	9.39	8.87	6.63	5.71	5.21	7.57	9.08	9.57	10.70
	HDD50	2973	783	645	409	117	13	0	0	0	1	78	299	628
	HDD65	6171	1248	1063	846	464	195	30	6	12	98	400	720	1089
	CDD50	3045	1	3	29	110	342	591	698	641	440	157	29	4
	CDD65	765	0	0	2	7	59	170	239	188	86	14	0	0
	CDH74	6739	0	0	17	68	543	1442	2220	1546	793	110	0	0
	CDH80	2071	0	0	2	6	145	461	766	426	246	19	0	0
(14) Wind	WSAvg	8.9	10.6	10.4	10.1	10.6	9.2	7.8	6.4	5.9	6.7	8.7	9.8	10.2
(15) Precipitation	PrecAvg	35.8	1.9	1.7	2.6	3.5	3.5	3.9	3.7	3.7	3.4	2.8	2.8	2.5
	PrecMax	53.0	5.7	5.4	6.0	6.9	8.3	9.5	9.8	11.7	10.0	7.5	6.0	5.3
	PrecMin	24.6	0.3	0.2	0.5	0.2	0.6	0.8	0.6	0.9	0.0	0.5	0.6	0.5
	PrecStd	5.7	1.2	1.1	1.2	1.5	1.7	2.0	2.0	2.1	2.0	1.7	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	61.3	76.2	80.8	89.0	92.3	95.0	90.8	91.0	82.7	70.2	60.9
		MCWB	53.9	53.3	63.0	63.3	70.4	72.3	76.6	75.0	72.1	69.0	58.1	57.0
	2%	DB	50.8	54.1	66.4	75.3	83.9	88.7	90.4	87.6	86.5	76.9	64.0	54.2
		MCWB	47.5	49.4	57.0	60.9	67.7	71.8	75.1	73.8	71.4	64.7	56.5	50.9
	5%	DB	45.1	47.3	60.8	71.6	80.4	85.3	87.6	84.4	82.1	72.3	60.1	48.3
		MCWB	41.6	43.6	53.1	58.6	66.4	70.2	73.3	71.9	68.8	62.2	53.9	45.2
	10%	DB	39.8	41.2	55.2	66.4	75.9	81.9	84.1	81.9	78.6	67.8	55.7	43.8
		MCWB	37.1	37.3	49.1	55.3	64.2	69.3	71.7	70.8	66.7	60.1	50.0	41.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	55.6	63.7	67.0	73.3	77.0	80.0	78.5	76.1	71.7	61.2	57.2
		MCDB	56.7	58.4	75.9	76.7	83.0	86.5	88.4	87.4	86.6	80.5	67.0	60.6
	2%	WB	48.2	49.7	58.9	63.1	70.6	74.9	77.7	76.2	73.5	67.4	58.0	50.9
		MCDB	49.9	53.8	65.6	71.2	80.0	84.0	86.3	83.9	82.3	72.4	62.5	53.7
	5%	WB	41.8	43.3	54.4	60.0	68.3	73.1	75.8	74.5	71.4	64.2	54.4	45.7
		MCDB	44.6	47.7	59.2	68.6	77.0	81.5	84.1	81.4	78.3	69.6	59.1	47.9
	10%	WB	37.1	37.7	49.3	57.1	66.1	71.3	74.0	72.7	69.2	61.0	50.8	41.4
		MCDB	39.7	41.1	55.3	65.9	73.7	78.9	81.7	79.2	75.6	67.1	55.8	44.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.5	17.5	21.0	20.2	19.4	20.3	20.2	21.5	18.9	15.4	12.5
		MCDBR	17.3	20.0	24.5	26.2	23.9	22.7	22.5	22.7	25.0	23.8	20.2	17.2
		MCWBR	15.2	16.5	16.3	15.0	12.5	9.9	10.0	10.8	11.6	13.3	14.7	15.2
	5% WB	MCDBR	16.3	19.7	22.0	22.9	20.6	18.4	19.3	18.5	20.4	19.9	18.2	16.0
		MCWBR	15.1	16.9	16.0	15.3	12.2	9.8	10.1	10.1	10.5	13.4	14.6	15.1
(40) Clear-Sky Solar Irradiance	taub		0.299	0.332	0.351	0.398	0.434	0.436	0.440	0.444	0.408	0.353	0.318	0.298
	taud		2.377	2.290	2.344	2.228	2.174	2.213	2.206	2.187	2.266	2.421	2.465	2.434
	Ebn at Noon		270	275	284	277	268	267	265	259	261	266	261	261
	Edh at Noon		26	33	36	43	46	45	45	44	38	29	24	23
(44) All-Sky Solar Radiation	RadAvg		474	740	1111	1465	1679	1903	1927	1682	1381	903	566	400
	RadStd		44	102	101	129	159	121	102	92	114	90	81	43

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	N/A
(47) Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page